ABSTRACT OF THE DISCLOSURE

A method for transmitting data encoded with a legacy, human-readable character set (e.g., UTF-8, UTF-16) over a high-cardinality, massively parallel communication interface. The method comprises: creating a specific encoding map that provides a one-to-one mapping between each character in a pre-defined legacy character set and a unique state of a single physical channel on the parallel interface. This allows for the transmission of one full character per channel per frame. This "retrofit" method enables existing text-based protocols and data formats to be tunneled over the new physical layer at extremely high speeds, treating entire characters as atomic units of transmission rather than sequences of bits.